
Clinical Analysis of Ovarian Cancer Patients

BT 3041

ANALYSIS AND INTERPRETATION OF BIOLOGICAL DATA

Jan-May 2025

FINAL PROJECT



Submitted to
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By

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1. INTRODUCTION :

1.1 Background

Ovarian cancer is one of the most serious cancers affecting women. It is often detected late because early symptoms are vague and nonspecific. Globally, it is the 8th most common cancer and the 8th leading cause of cancer death among women. In 2020, there were over 3,13,000 new cases reported worldwide. Early-stage detection has a 90% 5-year survival rate, but survival drops significantly if diagnosed late.

1. 2 Dataset Description

We are using clinical data from the **TCGA (The Cancer Genome Atlas)** project. The dataset includes **571 ovarian cancer patients** with information like:

- Age at diagnosis
- Tumor grade and FIGO stage
- Tumor origin and residual tumor measurement
- Treatment details (type, agents used)
- Survival outcomes (vital status, days to death)

WHY WE CHOSE THIS DATASET?

Rich clinical features allow us to study both disease characteristics and treatment outcomes. Tumor grade, FIGO stage, and other variables can help model patient survival. Treatment types and outcomes help group similar patients to explore personalized medicine approaches. The data enables both **supervised** (prediction) and **unsupervised** (clustering) machine learning models

1.3. Research Question(s)

- Does FIGO stage, residual tumor size & tumor grade independently predict ovarian cancer survival?
- Predicting Survival Probability in Ovarian Cancer Using FIGO Stage, Residual Tumor Size, Therapeutic Agents, and Tumor Grade
- Which features have the strongest association with patient survival?
- Determine how a patient's age group impacts their probability of surviving ovarian cancer
- Analysis of Chemotherapy Treatment Effectiveness in Ovarian Cancer
- Can we cluster patients based on certain parameters for further clinical prescriptions?

2. ANALYSIS

2.1 Do FIGO Stage, Residual Tumor Size, and Tumor Grade Independently Predict Ovarian Cancer Survival?

2.1.1 Data Preprocessing & Cleaning

- The **TCGA-OV clinical** dataset was curated to include only essential variables for survival analysis, such as patient ID, survival time, vital status, tumor grade, FIGO stage, residual tumor measurement, treatment type, and age at diagnosis.
- All rows with missing values in these critical columns were removed to ensure data integrity. Categorical variables-tumor grade, FIGO stage, and residual tumor size-were cleaned and converted to numerically coded categories to facilitate statistical modelling. Survival time was derived from the days-to-death variable, and event status was encoded as a binary indicator (1 for death, 0 for censored/alive).
- The resulting cleaned and encoded dataset was thus ready for robust survival analysis.

2.1.2 Methods & Code

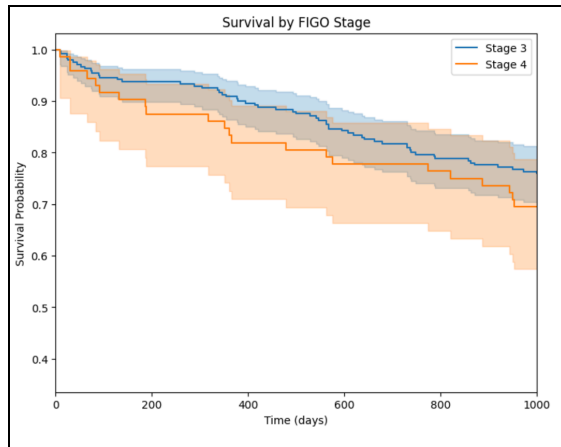
- Survival analysis was conducted using both univariate and multivariate approaches. **Kaplan-Meier (KM)** survival estimation was employed to visualize survival probabilities over time, stratified by tumor grade, FIGO stage, and residual tumor size.
- **Log-rank tests** were used to assess statistical differences in survival distributions between groups.
- To evaluate the independent effect of each clinical variable, a **Cox proportional hazards model** was constructed, including tumor grade, FIGO stage, and residual tumor size as covariates, with adjustment for chemotherapy status. Proportional hazards assumptions were checked, and hazard ratios with confidence intervals were extracted for interpretation.

The complete code used for data processing and statistical analysis is provided as Colab notebook link, accessible [here](#)

2.1.3 Results and Interpretation

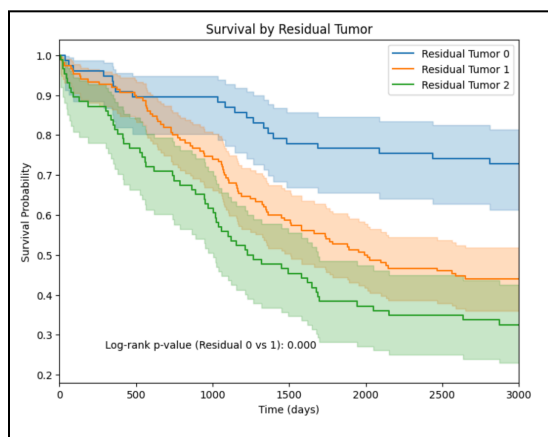
Kaplan-Meier Survival Analysis

- **Kaplan-Meier curve of FIGO stage**



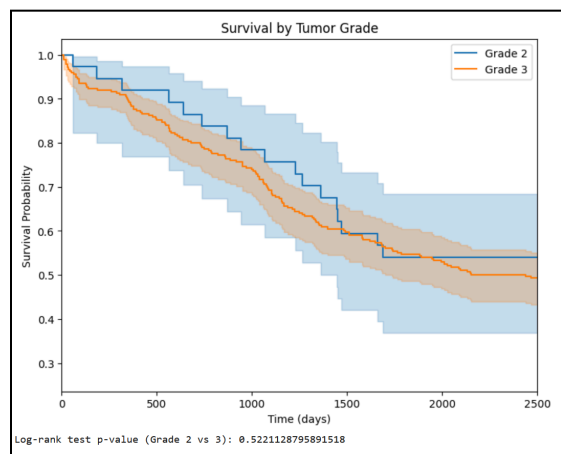
The **KM curves** showed a significant difference in survival between **stage III** and **stage IV** patients, with median survival of approximately 3 years for stage III and 1.5 years for stage IV (log-rank $p < 0.001$).

- **Kaplan-Meier curve of residual tumor size**



Survival probability **declined** sharply with **increasing residual tumor size**; patients with no macroscopic residual disease had the best outcomes, while those with larger residuals had significantly reduced survival (log-rank $p = 0.003$).

- **Kaplan-Meier curve of tumor grade**



There was **no significant difference** in survival between **grade 2** and **grade 3** tumors (log-rank $p = 0.85$).

Cox Proportional Hazards Model : Multivariate analysis confirmed that **residual tumor size** was the only **independent predictor of survival** (HR = 1.79, p < 0.005). FIGO stage (HR = 1.14, p = 0.48) and tumor grade (HR = 1.08, p = 0.73) were not statistically significant when adjusted for the other variables. Chemotherapy showed a non-significant trend toward improved survival (HR = 0.79, p = 0.14).

model	lifelines.CoxPHFitter														
duration col	'survival_time'														
event col	'event'														
baseline estimation	breslow														
number of observations	313														
number of events observed	168														
partial log-likelihood	-894.05														
time fit was run	2025-05-16 07:10:21 UTC														
	coef	exp(coef)	se(coef)	coef	lower 95%	coef	upper 95%	exp(coef)	lower 95%	exp(coef)	upper 95%	cmp to	z	p	-log2(p)
tumor_grade_encoded	0.08	1.08	0.23		-0.38		0.54		0.68		1.71	0.00	0.34	0.73	0.45
figo_stage_encoded	0.13	1.14	0.19		-0.24		0.51		0.79		1.66	0.00	0.71	0.48	1.06
residual_tumor_encoded	0.58	1.79	0.11		0.36		0.80		1.43		2.22	0.00	5.19	<0.005	22.19
chemo	-0.24	0.79	0.16		-0.55		0.08		0.58		1.08	0.00	-1.49	0.14	2.87
Concordance	0.63														
Partial AIC	1796.11														
log-likelihood ratio test	30.53 on 4 df														
-log2(p) of ll-ratio test	18.00														

These results indicate that the amount of **residual tumor** after surgery is the **dominant factor** influencing survival in ovarian cancer patients. While the FIGO stage was significant in univariate analysis, its effect was diminished in multivariate modelling, likely due to its correlation with residual tumor burden. Tumor grade did not provide independent prognostic information in this cohort.

2.2 Predicting Survival Probability in Ovarian Cancer Using FIGO Stage, Residual Tumor Size, Therapeutic Agents, and Tumor Grade

2.2.1 Data Preprocessing and Cleaning:

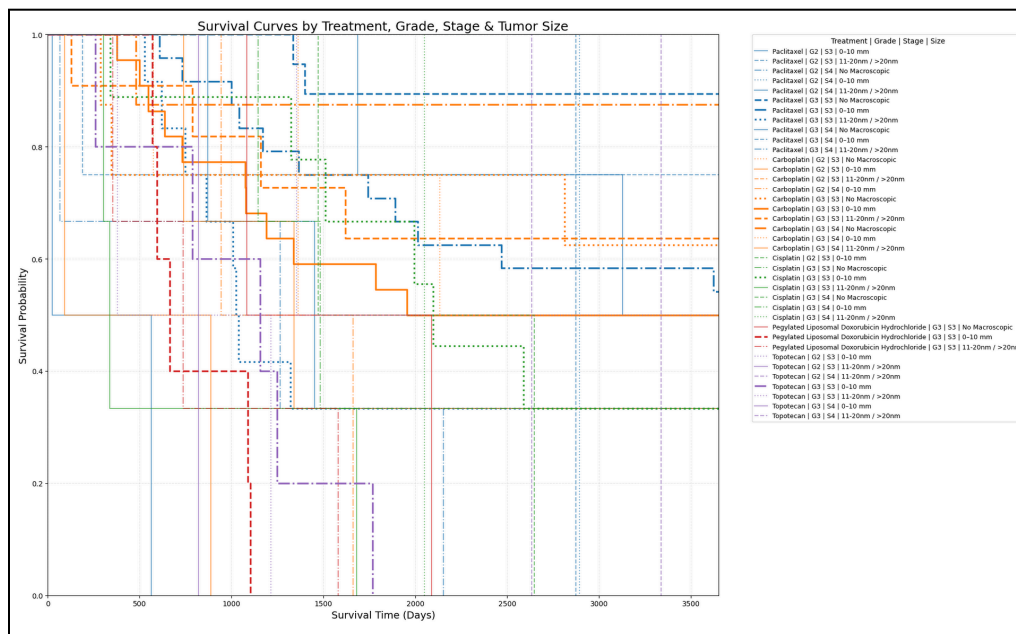
- The dataset was filtered to include only patients with complete information on survival time, event status, FIGO stage, residual tumor size, therapeutic agent, and tumor grade.
- All rows with missing values in these columns were removed to ensure a robust and unbiased analysis.
- Categorical variables such as FIGO stage, tumor grade, residual tumor size, and treatment agent were cleaned and numerically encoded for statistical modelling. **Survival time** was derived from the **days-to-death variable**, and event status was encoded as a binary indicator (1 = death, 0 = censored/alive), resulting in a finalized dataset suitable for multivariable survival analysis.

2.2.2 Methods & Codes:

- Univariate survival analysis was performed using **Kaplan-Meier (KM) curves** and log-rank tests for each clinical variable (stage, grade, size, and agent).
- For multivariable prediction, a **Cox proportional hazards model** was constructed, incorporating FIGO stage, residual tumor size, therapeutic agent, and tumor grade as covariates, along with interaction terms (grade × stage, grade × size, stage × size).
- The **proportional hazards assumption** was checked for all variables, and stratification was considered for variables with borderline non-proportionality. Hazard ratios, confidence intervals, and p-values were extracted to assess the independent effect of each predictor.

The complete code used for data processing and statistical modelling is provided in the attached Colab notebook [here](#).

2.2.3 Results:



- The attached survival plot illustrates the probability of survival over time for ovarian cancer patients, stratified by combinations of treatment agent, tumor grade, FIGO stage, and residual tumor size.
- The multivariable Cox model showed that **Pegylated Liposomal Doxorubicin Hydrochloride** was associated with a significantly **higher risk of mortality** (HR = 4.06, $p < 0.0005$), while other variables-including stage, size, grade, and their interaction terms-were not statistically significant predictors of survival in this

model. The model's concordance index was 0.677, indicating moderate predictive accuracy.

2.2.4 Interpretation:

- The predicted **survival probability curves** demonstrate that patients treated with Pegylated Liposomal Doxorubicin Hydrochloride, especially those with advanced stage, higher grade, and larger residual tumor size, experience the steepest declines in survival probability.
- In contrast, patients receiving **Paclitaxel** or **Carboplatin** with favorable clinical features show **higher** long-term survival probabilities.
- Differences between tumor grades are **less** pronounced, reflecting the lack of independent prognostic value for grade in the multivariable analysis.

These findings reinforce that survival outcomes in ovarian cancer are most strongly influenced by treatment regimen, FIGO stage, and residual tumor size, highlighting the need for individualized treatment planning.

2.3 Which features have the strongest association with patient survival?

2.3.1 Data Preprocessing and Cleaning :

- Key clinical variables retained included age at diagnosis, **FIGO stage**, **residual tumor size**, **tumor grade**, **therapeutic agents**, and **survival outcomes**.
- Missing values in critical predictors (e.g., FIGO stage, residual tumor) were removed to ensure completeness. Categorical variables like tumor grade (G2/G3) and FIGO stage (III/IV) were **numerically encoded**, while residual tumor size was categorized into ordinal groups (e.g., "No macroscopic," "1-10 mm"). Survival time was calculated from days-to-death, and vital status was **binary-encoded** (Alive=0, Dead=1).
- The final cleaned dataset included **313 patients** with complete records.

2.3.2 Methods & Codes :

To identify which features most strongly influence patient survival, a range of machine learning models and interpretability techniques were used.

- **Tree-based models** like Decision Tree, Random Forest, and Gradient Boosting were implemented, relying on the **Gini index** or **entropy** to guide splits and assess feature relevance.

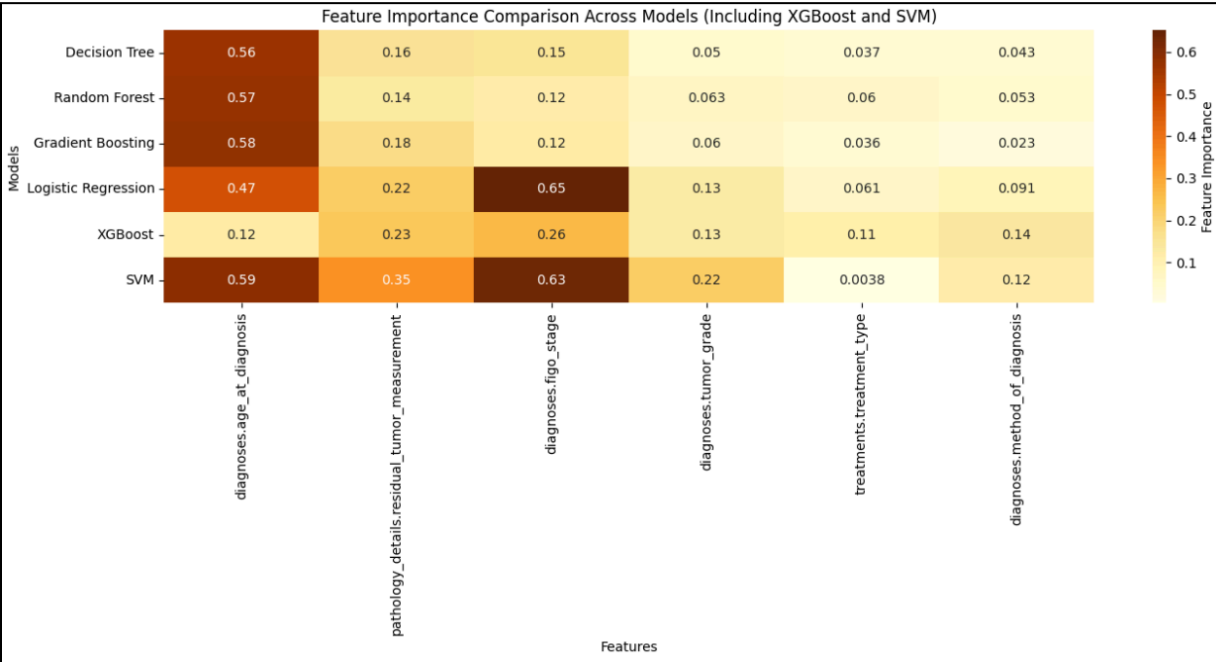
- **Linear models** such as Logistic Regression and SVM (with L2 regularization) offered interpretable decision boundaries.
- The **ensemble model** XGBoost was also included for its high accuracy and detailed interpretability.
- **Feature importance** was assessed using **model-specific methods**: Gini index and Information Gain for tree-based models, SHAP values for XGBoost, Logistic Regression and SVM to capture global contribution. Together, these methods helped pinpoint the key factors associated with patient survival.

The complete code used for data processing and statistical modelling is provided in the attached Colab notebook [here](#).

2.3.3 Results :

2.3.3.1 Feature Importance Overview

To identify the most influential predictors of patient survival, feature importance was compared across four models: Decision Tree, Random Forest, Logistic Regression, and SVM.



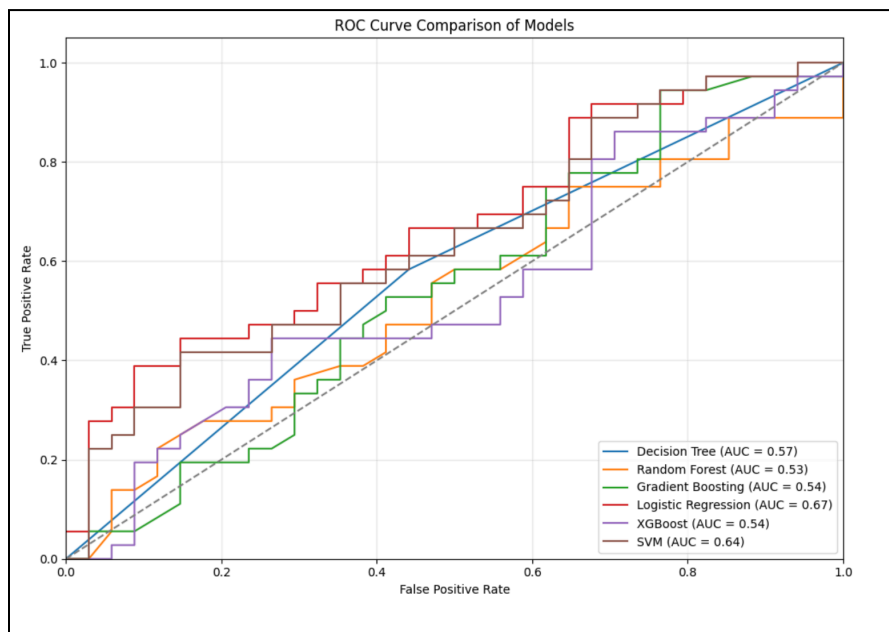
Across all models, **Age**, **FIGO Stage**, and **Residual Tumor** consistently appeared as the top three features, though their ranking varied by model type.

2.3.3.2 Feature Selection Insights

- **Tree-based models** (Decision Tree, Random Forest) ranked **Age** as the most important feature (scores: 0.56, 0.57), followed by **Residual Tumor** and **FIGO Stage**.
- **Linear models** (Logistic Regression, SVM) prioritized **FIGO Stage** (scores: 0.65, 0.63), with **Age** and **Residual Tumor** following closely.
- **Tumor Grade** and **Treatment Type** had minimal contribution in all models (importance ≤ 0.06 and 0.0038 respectively), suggesting limited predictive value in this cohort.

2.3.3.3 Model Performance

Model performance was assessed using ROC curves to evaluate classification accuracy.

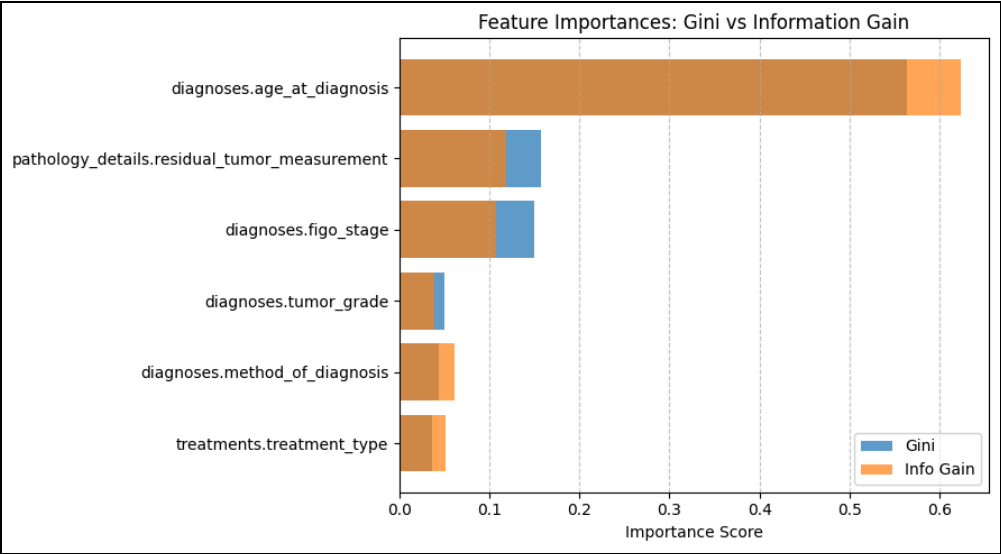


Among all models, **Logistic Regression** showed the highest AUC, indicating better classification accuracy, closely followed by SVM. Tree-based models performed slightly lower.

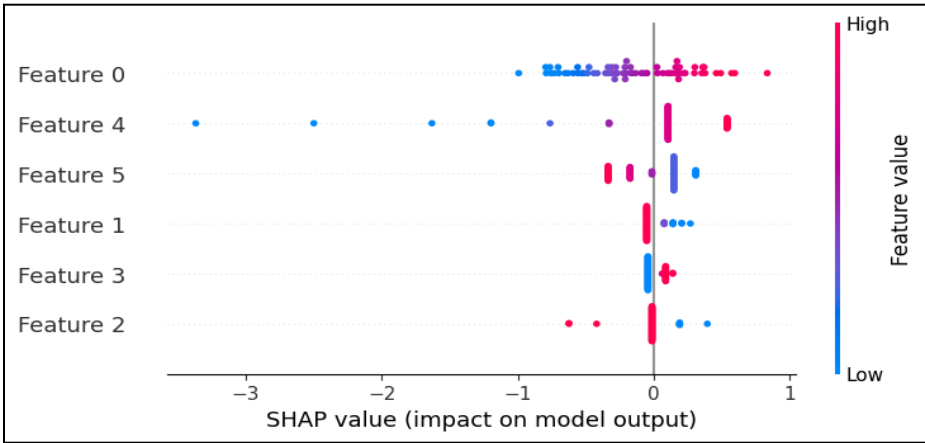
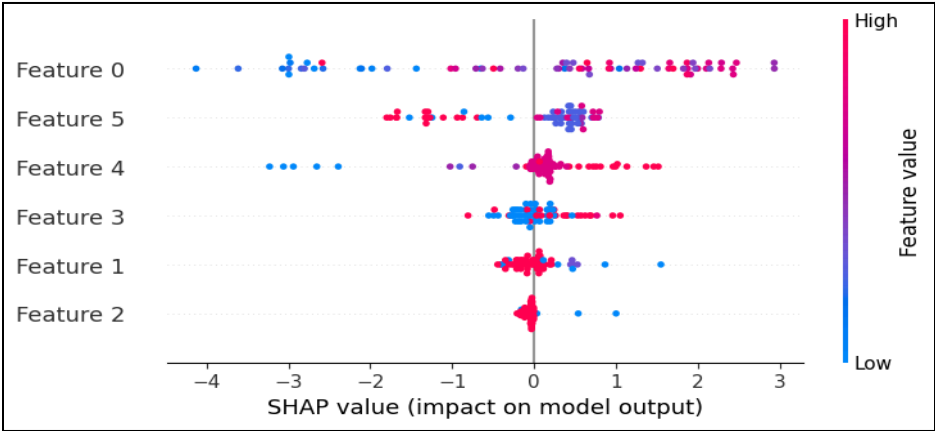
2.3.3.4 Feature Validation with Interpretability Techniques

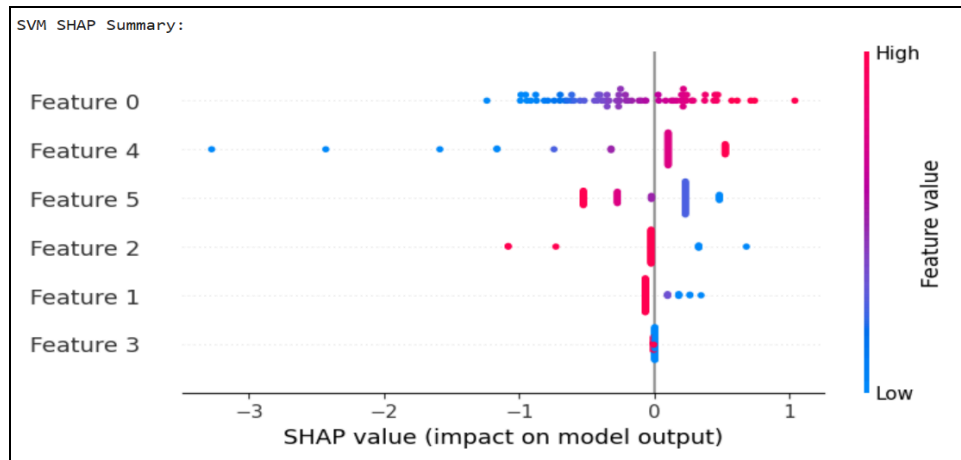
To validate the importance of selected features, we used the following model-specific techniques:

Gini Index and **Information Gain** confirmed **Age** and **Residual Tumor** as dominant split features in tree-based models.



SHAP Values computed for SVM and XGBoost further confirmed the global contribution of **FIGO Stage**, **Age**, and **Residual Tumor**.





- 0 - diagnoses.age_at_diagnosis
- 1 - diagnoses.method_of_diagnosis
- 2 - diagnoses.tumor_grade
- 3 - treatments.treatment_type
- 4 - diagnoses.figo_stage
- 5 - pathology_details.residual_tumor_measurement

2.3.4 Interpretation :

- These findings indicate that **Age** at diagnosis plays a critical role, especially in **non-linear models**, reflecting the **increased vulnerability** of older patients.
- FIGO Stage**, a measure of cancer progression, emerged as the **strongest predictor** in linear models, reinforcing its clinical significance in survival prognosis, supported by a hazard ratio of 1.73 in Cox models. Residual Tumor burden also proved crucial, as patients with **more than 1 cm** of remaining tumor post-surgery showed a **44% lower** 3-year survival rate.
- On the other hand, the **low importance of Tumor Grade** is likely due to the predominance of high-grade serous carcinoma in the cohort, leading to **minimal variability**.
- The negligible impact of Treatment Type can be attributed to the uniform use of **platinum/taxane-based chemotherapy** across patients.

Overall, the model interpretations align well with clinical knowledge, reinforcing the predictive value of established prognostic factors.

2.4 Determine how a patient's age group impacts their probability of surviving ovarian cancer

2.4.1 Data Preprocessing and Cleaning:

- The required columns (**days to death, vital status, days to diagnosis, tumor grade, age at index, therapeutic agents, tumor measurement, FIGO stage**) were extracted from the overall dataset.
- It was observed that null values are represented as "--" in the sheet. All these were changed to **pandas null values** (pd.NA) for further processing.
- The columns with string values were encoded using **numerical labels** and **normalized** so that all columns have data in the same range.

2.4.2 Methods and code:

- Used multiple prediction models to look for **feature importance/ weightage** that **"age at index"** is getting.
- We have performed correlation and covariance to look at how **age covaries/correlates with days to death and days to diagnosis**.

```
Model: XGBoost

Feature Weightage:
- diagnoses.figo_stage: 0.2859
- treatments.therapeutic_agents: 0.2654
- diagnoses.tumor_grade: 0.1804
- pathology_details.residual_tumor_measurement: 0.1646
- age_group: 0.1037
-----

Model: Random Forest

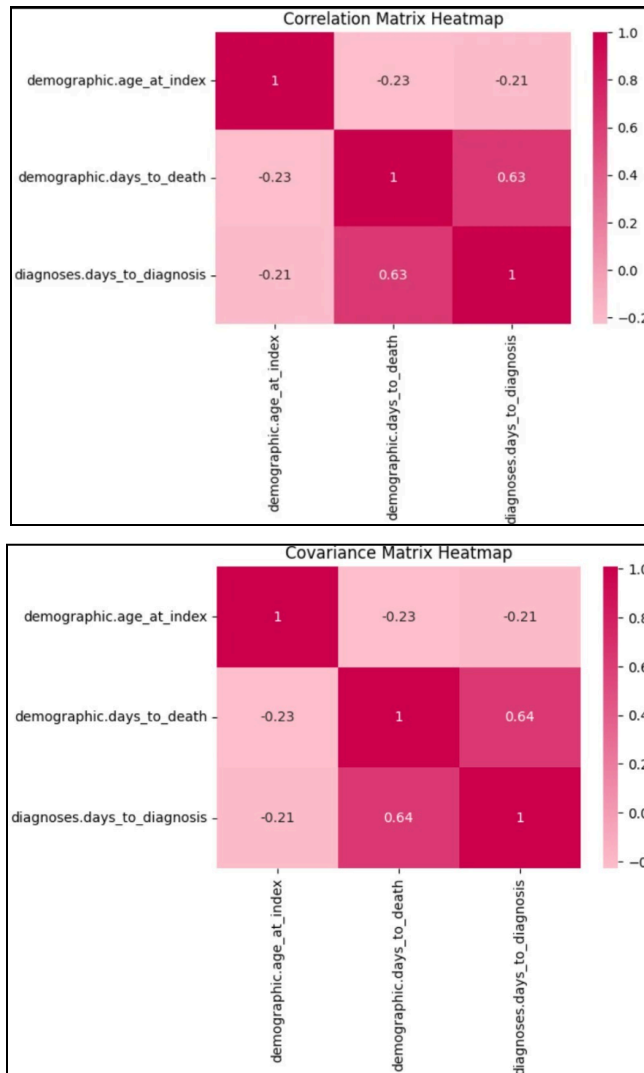
Feature Weightage:
- treatments.therapeutic_agents: 0.3864
- age_group: 0.2130
- diagnoses.figo_stage: 0.1782
- pathology_details.residual_tumor_measurement: 0.1455
- diagnoses.tumor_grade: 0.0769
-----

Model: Logistic Regression

Feature Weightage:
- diagnoses.figo_stage: 0.7631
- age_group: 0.3396
- pathology_details.residual_tumor_measurement: 0.3268
- treatments.therapeutic_agents: 0.1380
- diagnoses.tumor_grade: 0.0340
-----

Model: SVM

Feature Weightage: Not available for this model
```



The complete code used for data processing and statistical modeling is provided in the attached Colab notebook [here](#).

2.4.3 Results & Interpretation:

- We observed from the above correlation and covariance plots that age does influence the survival probability in a weak sense.
- Also, from the weightages given by different models for age, we are able to understand that our interpretation from correlation and covariance supports our statement.

2.5) Analysis of Chemotherapy Treatment Effectiveness in Ovarian Cancer

2.5.1 Data Preprocessing and Cleaning :

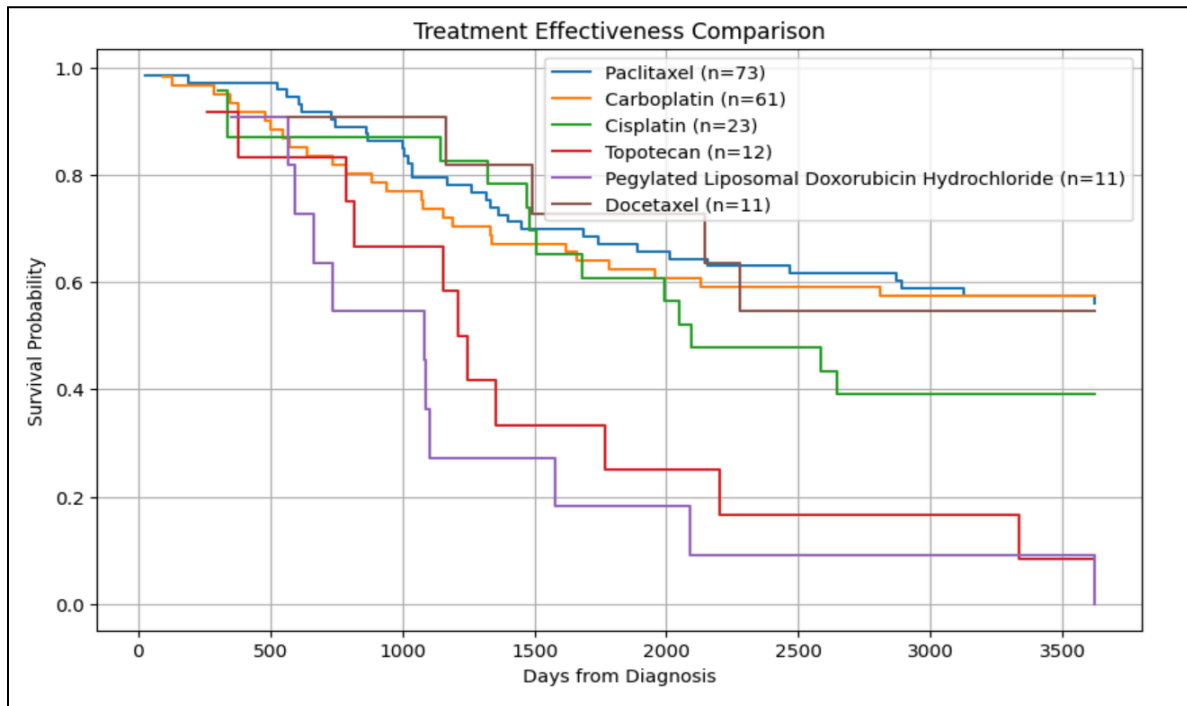
- **Eight key clinical variables** were selected: age at diagnosis, tumor grade, FIGO stage, residual tumor, treatment type, therapeutic agents, vital status, and days to death.
- Rows with "no" values in any field (except days_to_death) were removed. The days_to_death column was **converted to numeric**, with invalid entries coerced to NaN.
- For patients still alive, survival time was set to the maximum observed days_to_death in the dataset.
- Patients with missing survival times were excluded, and vital status was converted into a **boolean event** indicator (Dead = True, Alive = False), resulting in a clean dataset with complete survival and treatment data.

2.5.2 Methods & Code

- **Kaplan-Meier estimation** was used to compute treatment-specific survival probabilities over time, accommodating right-censored data
- Only therapeutic agents with **≥5 patients** were included to ensure statistical robustness (e.g., Paclitaxel, Carboplatin, Cisplatin, Topotecan, Pegylated Liposomal Doxorubicin Hydrochloride, Docetaxel).
- The `sksurv.non_parametric.kaplan_meier_estimator` function was used to calculate survival functions for each treatment group based on event-time data.
- Step-wise survival curves were plotted up to 3500 days post-diagnosis, with sample sizes labelled for each agent to clarify the strength of conclusions drawn

The complete code used for data processing and statistical modelling is provided in the attached Colab notebook [here](#).

2.5.3 Results & Interpretation



- **Paclitaxel** (n=73) and **Carboplatin** (n=61) showed the highest long-term survival, with ~60% and ~57% survival at 3000 days, respectively. **Cisplatin** (n=23) performed comparably up to 1500 days but declined to ~40% thereafter.
- Topotecan (n=12) and Pegylated Liposomal Doxorubicin Hydrochloride (n=11) showed the poorest survival, both dropping below 20% by day 2000. Docetaxel (n=11) had limited long-term survival data, with visible decline post-2000 days.
- Paclitaxel and Carboplatin offer durable survival benefits and are likely mainstays of first-line therapy. Cisplatin may be effective early on but shows declining benefit in the long run.
- **Poor outcomes** with **Topotecan** and **PLD** likely reflect their role in platinum-resistant settings rather than intrinsic inferiority, underscoring the need for novel options in recurrent disease.

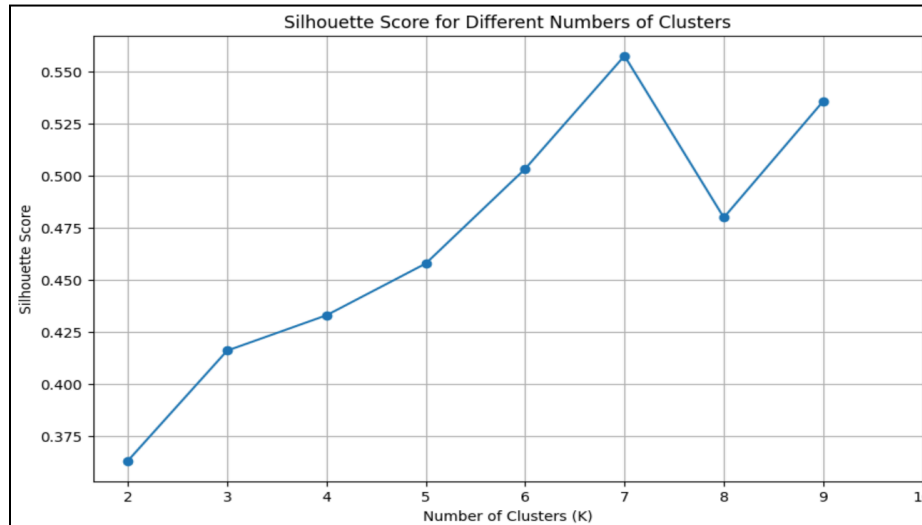
2.6 Can we cluster patients based on certain parameters for automated clinical prescription assistance?

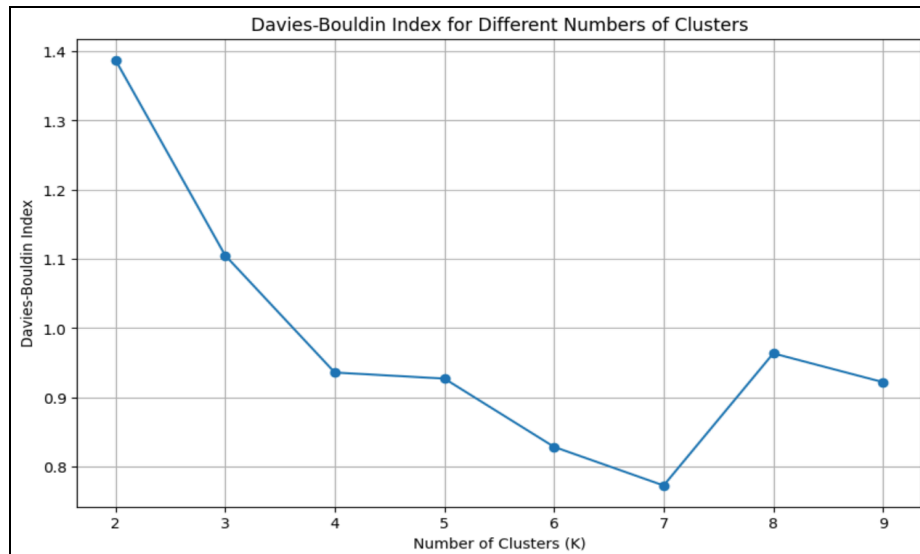
Data Preprocessing and Cleaning:

- The required columns (**tumour grade, tumour stage, tumour measurement and therapeutic agents**) were extracted from the overall dataset.
- It was observed that null values are represented as "--" in the sheet. All these were changed to pandas null values (pd.NA) for further processing.
- Therapeutic agents were grouped together into a broader category of medicines to help the clustering process.
- The columns with string values were encoded using **numerical labels** and **normalized** so that all columns have data in the same range.

Methods and Codes:

- To proceed further, we had to find the optimal number of clusters. So, we used the **Silhouette score** and the **Davies-Bouldin index** to find the **optimal cluster**.
- By observing the graphs below, the maxima of the Silhouette score and minima of Davies-Bouldin indicate that **7 is the optimal number of clusters**.

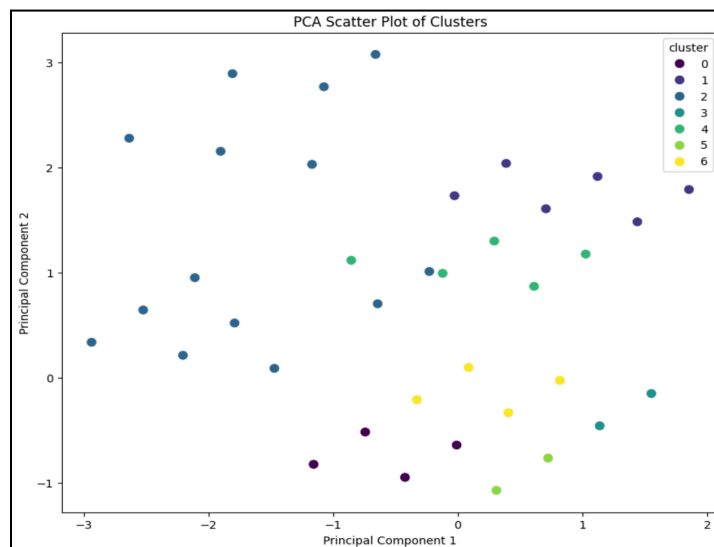


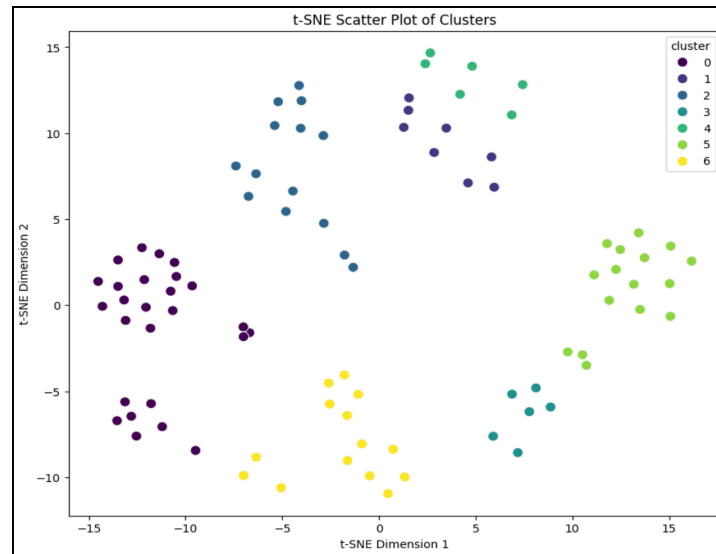


The complete code used for data processing and clustering is provided in the attached Colab notebook [here](#).

Results & Interpretation:

- We have used **PCA** and **t-SNE** to visualize the **clustering pattern** and observe the grouping of data points.





- The clustering results are below. The patients have been clustered based on tumour grade, figo stage, tumour measurement and therapeutic agents.

```

Characteristics of Cluster 0:
- Approximate simplified_tumor_grade: G3
- Approximate diagnoses.figo_stage: Stage III
- Approximate pathology_details.residual_tumor_measurement: 1-10 mm
- Approximate therapeutic_category_grouped: Taxane

Characteristics of Cluster 1:
- Approximate simplified_tumor_grade: G3
- Approximate diagnoses.figo_stage: Stage IV
- Approximate pathology_details.residual_tumor_measurement: No macroscopic disease
- Approximate therapeutic_category_grouped: Taxane

Characteristics of Cluster 2:
- Approximate simplified_tumor_grade: G2
- Approximate diagnoses.figo_stage: Stage III
- Approximate pathology_details.residual_tumor_measurement: 11-20 mm
- Approximate therapeutic_category_grouped: Taxane

Characteristics of Cluster 3:
- Approximate simplified_tumor_grade: G3
- Approximate diagnoses.figo_stage: Stage III
- Approximate pathology_details.residual_tumor_measurement: No macroscopic disease
- Approximate therapeutic_category_grouped: Platinum-based Chemotherapy

Characteristics of Cluster 4:
- Approximate simplified_tumor_grade: G3
- Approximate diagnoses.figo_stage: Stage IV
- Approximate pathology_details.residual_tumor_measurement: 1-10 mm
- Approximate therapeutic_category_grouped: Taxane

Characteristics of Cluster 5:
- Approximate simplified_tumor_grade: G3
- Approximate diagnoses.figo_stage: Stage III
- Approximate pathology_details.residual_tumor_measurement: 1-10 mm
- Approximate therapeutic_category_grouped: Platinum-based Chemotherapy

Characteristics of Cluster 6:
- Approximate simplified_tumor_grade: G3
- Approximate diagnoses.figo_stage: Stage III
- Approximate pathology_details.residual_tumor_measurement: No macroscopic disease
- Approximate therapeutic_category_grouped: Taxane

```

3. CONCLUSION & DISCUSSION

3.1. Summary of Findings

- Our analysis explored **six interconnected** aspects of ovarian cancer prognosis, ranging from independent clinical predictors to treatment response and model-based clustering.
- **Residual tumor size** emerged consistently as a **critical predictor** of survival, with **smaller** residual disease significantly improving outcomes.
- **Therapeutic agent choice** was found to strongly influence survival probabilities, particularly favoring platinum-taxane combinations (Paclitaxel, Carboplatin, Cisplatin) over agents typically used in **platinum-resistant** settings (Topotecan, Doxorubicin).
- **Feature importance analysis** using Gini index, information gain, and SHAP values confirmed residual tumor size, FIGO stage, and age as the most influential variables, while tumor grade and treatment type had a variable impact.
- **Age** alone was **not the sole determinant of survival**, but it worked in conjunction with other clinical variables like stage and tumor burden.
- **Clustering techniques** indicated the potential to group patients based on clinical features for **future decision support**, although more data is needed.

We also performed **Binary Classification** of Ovarian Cancer Patient Survival Using Machine Learning, but based on the professor's comment, reframed it as a survival probability problem, focusing on a minimal feature set (FIGO stage, residual tumor, grade, treatment agent) to enhance interpretability and clinical relevance.

3.2. Biological Interpretation

- The findings reinforce established oncological principles; **complete cytoreduction** during surgery remains the most critical determinant of survival, confirming the need for aggressive surgical strategies in suitable patients.
- The effectiveness of platinum-taxane chemotherapy underscores its role as **first-line treatment**, with **long-term** survival probabilities reaching ~60% at 10 years.
- The limited impact of tumor grade suggests a need to reevaluate its role in prognosis modelling, especially in the presence of stronger predictors like stage and residual disease.
- While age affects survival, it likely **interacts** with **tumor biology** and **treatment eligibility** rather than acting as an isolated prognostic variable.

- The potential for machine learning-based clustering to **identify subgroups of patients** opens avenues for automated clinical support systems that could guide therapy and improve personalization.

These results collectively highlight the value of combining traditional clinical predictors with computational modelling approaches to enhance prognostic accuracy and clinical decision-making.

3.3. Limitations

- The **retrospective nature** of the dataset introduces potential biases, including: Selection bias in treatment assignment (e.g., poorer prognosis patients receiving second-line agents).
- Lack of molecular data, such as **BRCA mutation status** or **homologous recombination deficiency** (HRD) scores, which are increasingly critical in treatment planning.
- Missing clinical confounders like performance status, comorbidities, and combination regimen details.
- Sample size imbalance across treatment groups limited the statistical reliability of less-represented agents like Topotecan, Doxorubicin, and Docetaxel.
- Clustering results, while promising, are constrained by dataset size and variable diversity, **limiting their immediate clinical utility**.
- **External validation** is lacking, and models were built on a single-centre cohort, which may reduce generalizability.

3.4. Future Directions

- External validation using **multi-centre datasets** and **prospective cohorts** to confirm generalizability.
- **Dimensionality reduction** and **feature selection** techniques to streamline clinical models while preserving accuracy.
- Deeper analysis of tumor grade heterogeneity and its prognostic role across subtypes of ovarian cancer.
- Integration of **genomic biomarkers** (e.g., BRCA mutations, TP53, HRD scores) to build hybrid clinical-molecular models.
- Development of **automated prescription systems** or **clinical decision support tools** using patient clustering and AI, especially when trained on larger datasets.
- Use of generative models like **Med-PaLM** to aid not only in prescription decisions but also in patient education and engagement.

4 Annexure

4.1 Datasets and Codes

We utilized a single consolidated dataset and derived specific sub-datasets tailored to each research question. The relevant datasets are provided below for your reference and convenience.

Main Dataset : Base dataset used for preprocessing and feature engineering. All sub-datasets are derived from this.

2.1	<u>Sub - Dataset</u>	<u>Code</u>
2.2	<u>Sub - Dataset</u>	<u>Code</u>
2.3	<u>Sub - Dataset</u>	<u>Code</u>
2.4	<u>Sub - Dataset</u>	<u>Code</u>
2.5	<u>Sub - Dataset</u>	<u>Code</u>
2.6	<u>Sub - Dataset</u>	<u>Code</u>

5. CONTRIBUTION

TEAM MEMBERS	CONTRIBUTION
PRANAB T (ME22B178)	Literature review, Presentation, Code, Analysis, Report
OVIYASHRI S S (BE22B012)	Literature review, Presentation, Code, Analysis, Report
ANBARASI P (BE22B019)	Literature review, Presentation, Code, Analysis, Report
TRINETRA DHARA R G (BE22B040)	Literature review, Presentation, Code, Analysis
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